Stanford De-escalation Guide for Gram-negative Bacteremia

Antibiotic Selection

<table>
<tr><th>Pathogens</th><th>Preferred therapeutic options IF SUSCEPTIBLE
Switch to PO when clinically stable, able to take orals, no concern for absorption issues</th></tr>
<tr><td rowspan="2">E.coli
Klebsiella spp.
Proteus spp.
Citrobacter koseri</td><td><u>Preferred:</u>
• Cefazolin 2g IV q8h
• Ceftriaxone 2g IV q24h
• Ciprofloxacin 500-750mg* PO BID
• Levofloxacin 500-750mg* PO daily
• TMP-SMX 2DS PO BID or 8-10mg TMP/kg/day PO divided in 2 or 3 doses

<u>Oral alternatives if unable to take preferred agents†:</u>
• Cephalexin 1g PO 4x daily
• Amoxicillin 1g PO TID if ampicillin susceptible
• Amoxicillin/clavulanate 875/125mg PO TID‡ if ampicillin susceptible and broader spectrum desired. Call Micro Lab (x48632) to request susceptibility testing if ampicillin (sulbactam)-intermediate or resistant.
• If no other options, cefpodoxime may be considered when at the tail end of treatment and monitored closely.† Low likelihood of achieving target exposures.
• Avoid cefdinir- poor clinical outcomes observed.[21]

† In uncomplicated bacteremia, data indicate a 1.5-5.5% risk of infection recurrence with typical doses of oral beta-lactams. This risk may be mitigated by administering oral beta-lactams after 3-5 days of active upfront parenteral or preferred therapy, use of amoxicillin or amoxicillin/clavulanate only in isolates with ampicillin or amoxicillin/clavulanate MIC ≤ 2 mg/L, respectively.[13,22,24]
‡Alternatives: Amoxicillin/clavulanate 2g XR PO BID (if covered by insurance). If GI upset with 875/125 mg PO TID regimen may consider, with appropriate counseling, 875/125mg PO BID (qAM and qPM) + amoxicillin 1000mg q noon.</td></tr>
<tr><td><u>ESBL-producers</u>
Often ceftriaxone resistant + cefoxitin susceptible. At SHC, micro comments state these isolates “possibly harboring a cephalosporinase”
• Ertapenem 1g IV q24h
• Ciprofloxacin 500-750mg* PO BID
• Levofloxacin 500-750mg* PO daily
• TMP-SMX 8-10mg/kg/day PO divided in 2 or 3 doses
Note: Avoid most beta-lactams (including piperacillin-tazobactam and amoxicillin-clavulanate). May report as susceptible, but treatment failure may occur (MERINO trial)</td></tr>
<tr><td>Enterobacter cloacae, Klebsiella aerogenes, Hafnia alvei, Citrobacter freundii

(moderate-high risk AmpC production) [18,19**]</td><td>• Cefepime 2g IV q8h extended infusion
• Ertapenem 1g IV q24h
• Ciprofloxacin 500-750mg* PO BID
• Levofloxacin 500-750mg* PO daily
• TMP-SMX 8-10mg/kg/day PO divided in 2 or 3 doses
Note: Avoid ceftriaxone and piperacillin-tazobactam, even if reported as susceptible. Prolonged use may result in emergence of ceftriaxone resistance via selection of derepressed AmpC mutants (often ceftriaxone resistant + cefoxitin resistant).</td></tr>
<tr><td>Serratia marcescens,</td><td>• Same as above with additional options as follows:</td></tr>
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Morganella morganii, Providencia spp (low risk AmpC production).[18,19**]	• Ceftriaxone 2g IV q24h • Piperacillin-tazobactam 3.375-4.5g* IV q8h extended infusion
Pseudomonas aeruginosa	**Consider ID consult** • Cefepime 2g IV q8h extended infusion • Ceftazidime 2g IV q8h extended infusion • Piperacillin-tazobactam 4.5g* IV q8h extended infusion • Meropenem 1g IV q8h extended infusion • Ciprofloxacin 750mg PO BID • Levofloxacin 750mg IV/PO daily • Consult ID for multi-drug resistant strains and/or unable to take the above agents
Stenotrophomonas maltophilia	**ID consult recommended**
Acinetobacter baumanniii	**ID consult recommended.** Commonly resistant to many antibiotics. Ampicillin-sulbactam is usually active.

* Lower doses listed are for typical 70kg, normal renal function, tailored for the organism causing bacteremia. Higher dose may be considered for deep seated infections, obese (BMI ≥ 30), high CrCl > 100 ml/min. Use clinical judgement.

**Clinical reports of emergence of resistance has been reported mainly in *Enterobacter spp*[19] Higher mutation rates reported in experimental model of *E. cloacae complex, E, aerogenes, C. freundii, H. alvei* than *Providencia spp, Serratia spp. M. morganii.*[18]

Abbreviations: TMP-SMX= trimethoprim/sulfamethoxazole, DS = double strength, FQ= fluoroquinolone, PK/PD = pharmacokinetic/pharmacodynamic, MIC= minimum inhibitory concentration

Duration (excludes neutropenia- see FN pathway, consult ID**)**

Type	Duration of therapy	Notes
No evidence of deep-seated infection that would require prolonged treatment†	7 days[2, 7, 8, 25] Count day 1 from the 1st day of active therapy† ID consult if: • Patient is severely immunocompromised† • Considering a longer course of therapy	• Excludes neutropenic patients: see FN pathway; ID consult. • **Reminder**: Rule out infections involving long term catheters, ports, or hardware: longer treatment may be warranted if prosthesis/foreign materials are infected. Consider ID or ASP consult.
†Repeat blood cultures are generally not necessary to confirm clearance and are not necessary to determine day 1 of treatment.[10, 12] For clinically improved patients without source control issues, count day 1 from the 1st day of active therapy. Consult ID or ASP with additional questions or concerns.		

†BALANCE trial exclusions:

Patient has severe immune system compromise, as defined by: absolute neutrophil count <0.5x109/L; or is receiving immunosuppressive treatment for solid organ or bone marrow or stem cell transplant

Patient has a prosthetic heart valve or synthetic endovascular graft (post major vessel repair with synthetic material) (note: coronary artery stents are not an exclusion)

Patient has documented or suspicion of syndrome with well-defined requirement for prolonged treatment:
i) infective endocarditis;
ii) osteomyelitis/septic arthritis;
iii) undrainable/undrained abscess;
iv) unremovable/unremoved prosthetic associated
infection (e.g. infected pacemaker, prosthetic joint infection,
ventriculoperitoneal shunt infection etc.)

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